WAVEFORM GENERATOR QUICK REFERENCE (27 WAVES)

ID	Waveform Name	Function Name	Description
0	Sine	makeSine	Standard sine wave — 127 + 127·sin(2πi/N)
1	Triangle	makeTriangle	Linear triangle — 2·(i/N)−1 scaled 0–255
2	Saw Tooth	makeSaw	Rising sawtooth — 255·(i/N)
3	Square	makeSquare	50% duty square — (i < N/2) ? 255 : 0
4	EKG Wave	makeEKG	Heartbeat pulse — P-wave → QRS → T-wave
5	CRAZY!!!	makeCrazy1	Harmonic cocktail — sin(x)+sin(2x)/2+…
6	Stepped	makeSteps	Quantized staircase — floor(i/(N/S))·(255/(S−1))
7	Double Pulse	makeDoublePulse	Two narrow pulses — at N/4 and 3N/4
8	Random Noise	makeNoise	Pseudo-random — random(0,255)
9	Half Rect Sine	makeHalfRect	Half-rectified sine — max(0,sin()) scaled
10	BLOB!!	makeBlob	Raised cosine bump — 127 + 127·cos(2πi/N)
11	Bent Saw Tooth	makeBentSaw	Curved saw — 255·(i/N)^k
12	Shark Fin	makeSharkfin	Fast rise, instant drop — exponential rise
13	Hyper Sine	makeHyperSine	Clipped sine — clamp(A·sin())
14	Wobble Pulse	makeWobblePulse	AM pulse — square(x)·(0.5+0.5·sin())
15	Ripple Saw	makeRippleSaw	Saw + ripple — saw(i)+A·sin(k·2πi/N)
16	Folded Sine	makeFoldedSine	Wave-folded sine — sin(2.2πi/N) folded
17	SuperSaw	makeSuperSaw	Detuned saws — Σ detuned saws, normalized
18	SinFold	makeSinFold	Fold-distorted sine — sin(sin()·G)
19	PulseSweep	makePulseSweep	Sweeping duty — (i mod P < D(i)) ? 255 : 0
20	SpiralWave	makeSpiralWave	Phase-rotating sine — sin(x + k·sin(x))
21	ChaosSpark	makeChaosSpark	Chaotic bursts — logistic map + spikes
22	Formant	makeFormant	Vocal-like resonance — sin(x)·sin(F1x)·sin(F2x)
23	BitCrush Saw	makeBitCrushSaw	Bit-reduced saw — (saw >> B) << B
24	WindowPulse	makeWindowPulse	Windowed pulse — pulse(i)·hann(i)
25	TentMap	makeTentMap	Chaotic tent function — r·min(x,1−x)
26	FM Sweep	makeFMSweep	FM-modulated sweep — sin(x + m·sin(fm·x))

Waveform Reference TableN = buffer length, i = sample index (0…N-1), all outputs 0–255.

WIRING DIAGRAM (GRAYSCALE)

Arduino UNO R4 WiFi Connections

Buttons (Using Analog Pins as Digital Inputs)

- **A5 → Blue Button**
    - Manual Mode / Next Wave Tap to advance/Hold 1 sec, to return to Auto Mode
- **A3 → Yellow Button (Middle)**
    - Duty Adjust – Tap to adjust/Hold for fast adjust
- **A4 → Yellow Button (Right)**
    - Duty Adjust + Tap to adjust/Hold for fast adjust
    - Duty Mode Toggle (Double-tap)

All buttons share a **common ground bus**.

DAC Output

- DAC pin → Output jack / test point
- Ground → Output ground

LED Matrix

- Uses built-in UNO R4 LED matrix
- No external wiring required

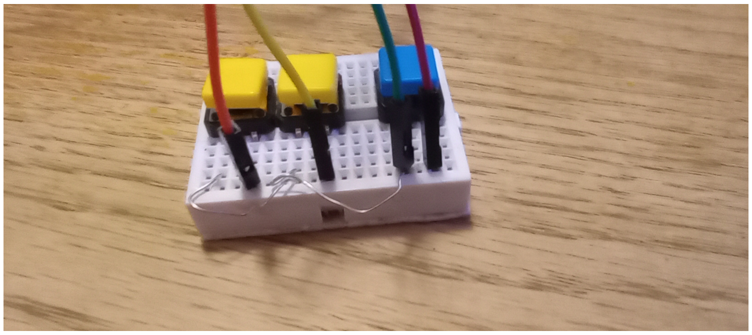
Power

- USB-C for power and programming
- 9v UNO power port

BUTTON LAYOUT DIAGRAM

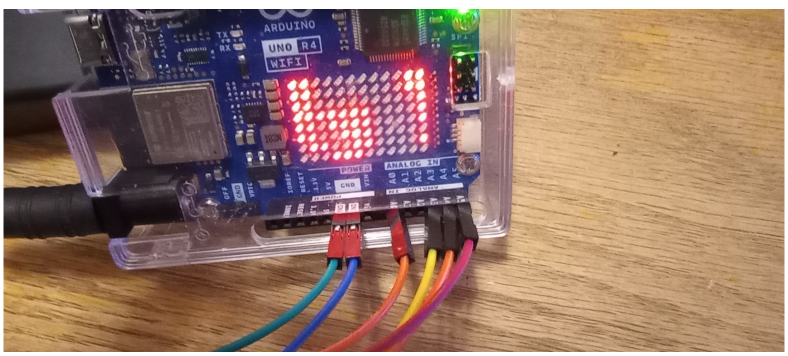
Breadboard Layout (Top-Down, Grayscale)

Code
```
[ Blue Button ]   [ Yellow – ]   [ Yellow + / Mode ]
      |              |               |
     A5             A3              A4
      |              |               |
--------------------------------------------------  ← Ground Bus
```



Button detail

- All buttons share the same ground wire
- Each button’s opposite leg goes to its respective UNO pin
- Internal pull-ups enabled (no resistors needed)



Arduino UNO R4 connections

From left to right

- Green GRND => Button assy. (Blue button ground bus)
- Blue GRND => Oscilloscope ground.
- Orange A0 => DAC output (Oscilloscope probe)
- Yellow A3 => Yellow button. Duty + / Duty toggle
- Orange A4 => Yellow Button. Duty -
- Red A5 => Blue button. Wave advance / automode-manual mode toggle

OPERATION GUIDE

Startup

- Device loads boot message
- Waveform 0 loads automatically
- DAC output begins immediately

Modes

Auto Mode

- Default mode
- Waveform scrolls continuously
- Advvances waveform every 5 sec
- Duty not shown unless duty mode is active

Manual Mode (Blue Button)

- Tap → Next waveform
- Manual mode indicator appears on matrix upper right corner
- Hold 1 sec, to return to Auto Mode

Duty Mode (Right (+) Yellow Button)

- **Single tap:** Adjust duty% 1 increment
- **Double tap:** Duty ON/OFF toggle
- **Duty ON:** Scroll shows “WaveName   Duty XX%”
- **Duty OFF:** Scroll shows only waveform name

Duty Adjust

- **Middle Yellow:** Duty (–)
- **Right Yellow:** Duty (+) / Duty ON/OFF toggle
- Hold buttons for fast adjust
- Live duty display appears immediately
- After timeout, scroll returns to combined Wave + Duty text

Scroll Behavior

- Waveform name scrolls continuously
- If duty is active, scroll includes duty percentage
- Live duty display overrides scroll temporarily
- Manual mode and duty mode icons appear in matrix corners

Troubleshooting

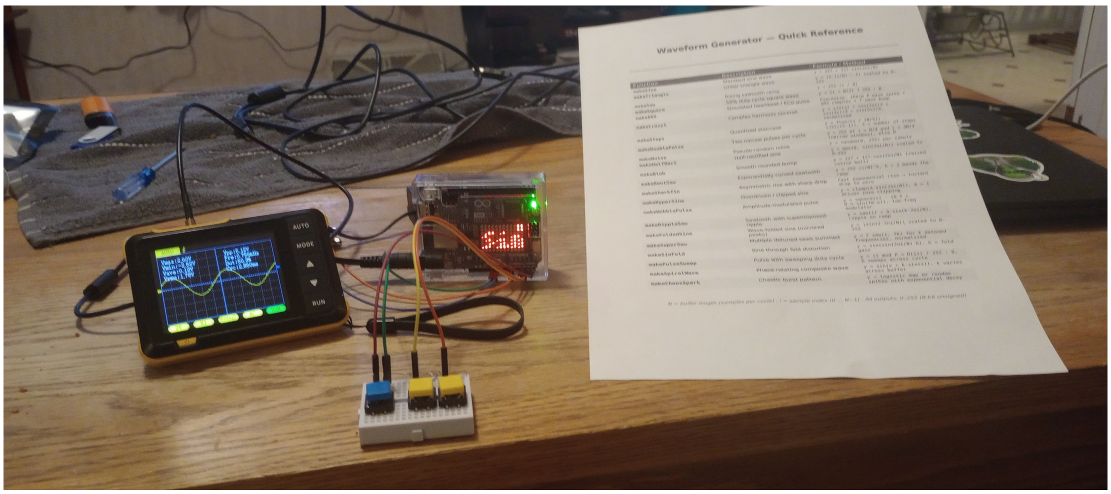
- **Blinking text:** Normal (matrix ignores alpha channel)
- **No fade effects:** Matrix does not support brightness levels
- **Duty not showing:** Duty mode must be ON
- **Wave changes hide duty:** Fixed in latest firmware

NOTES / VERSION

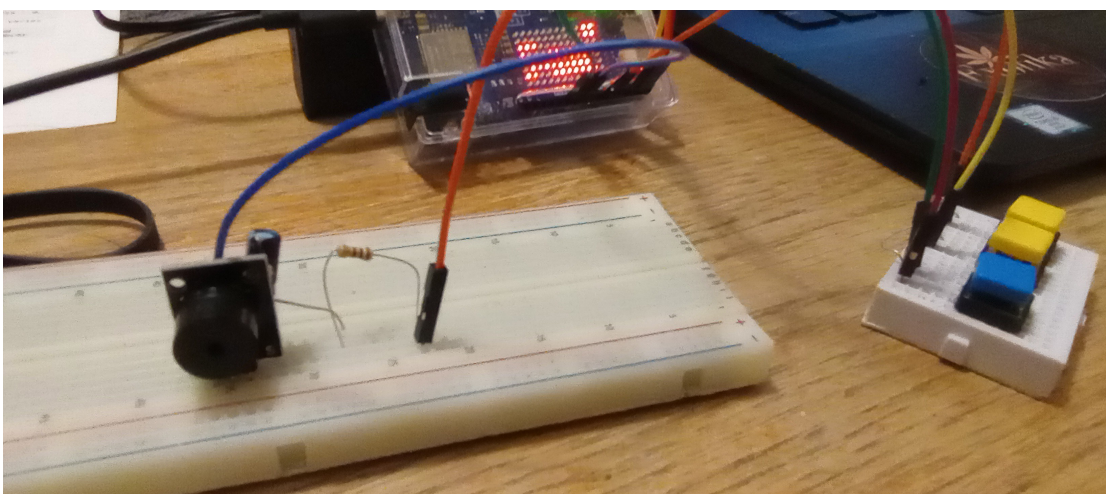
- Project: UNO R4 Waveform Generator
- Hardware: UNO R4 WiFi, LED Matrix, 3-button breadboard
- DAC: 8-bit output
- Firmware: Latest revision with duty-scroll fix

Notes:

- Add new waveforms here
- Add future features
- Space for wiring changes



Oscilloscope connected and running!



Optional passive buzzer hookup. (Hear your waveforms!)

You will need a *100 Ω resistor* *and* a ***10 µF electrolytic capacitor* (50v)**

1. On **10 µF electrolytic capacitor (50v)**, connect capacitor (–) side to "S" buzzer leg and (+) side to resistor.
2. Connect A0 – DAC output wire to 100 Ω resistor
3. GRND wire to – buzzer leg opposit "S" leg.(not the middle leg)
4. Kick back, enjoy the sounds!